



NEURAI PLATFORM

The AI-native operating surface for organizational memory

A complete product and architecture guide — beginning with Echo



Repository-grounded final edition · 26 August 2026

Architecture, product, trust, speech intelligence, market position, and expansion thesis

A final view—without pretending the future is already here

One document that separates product truth, binding architecture, current code, and expansion intent.

Snapshot discipline This edition is grounded in docs/SPEC.md, locked ARCHITECTURE.md decisions M1–M40, every tracked document under docs/, repository-wide route/schema/dependency inventories, and focused inspection of the live seams. It reflects commit 5c22477 plus the current local UI worktree on 26 August 2026.

Label	Meaning	How it is used here
Implemented	A route, data shape, UI surface, migration, or runtime path exists in the codebase.	Described as a present capability; external deployment health is not inferred.
Architecturally committed	A locked M-decision defines the required shape.	Described as the product contract even if a specific provider or rollout remains conditional.
Expansion path	The platform brief or AI-native plan points beyond Echo.	Presented as thesis/roadmap, never as shipped fact.

Source hierarchy. Product behavior conflicts resolve to SPEC.md. System-shape conflicts resolve to the locked architecture. Code demonstrates the current implementation; a UI control or draft note never silently overrules those contracts.

Naming. NeurAI Platform is the platform and shared shell. Echo is the call-intelligence app inside it. The Android recorder is Echo Mobile.

AI should be present before it is prompted

NeurAI Platform turns lived work into governed memory—and governed memory into useful action.

Most assistants begin after a person opens a chat and reconstructs context. NeurAI Platform begins where organizational truth is created: conversations, decisions, commitments, corrections, and follow-through. It captures that truth with provenance, makes it searchable across time, and lets an agent act only through the authority of the person who asked.

The north star Presence + hands + signals + bounded autonomy. The assistant sees what the person is already allowed to see, can operate the same product controls the person can operate, notices meaningful events, and asks before consequences when policy requires it.

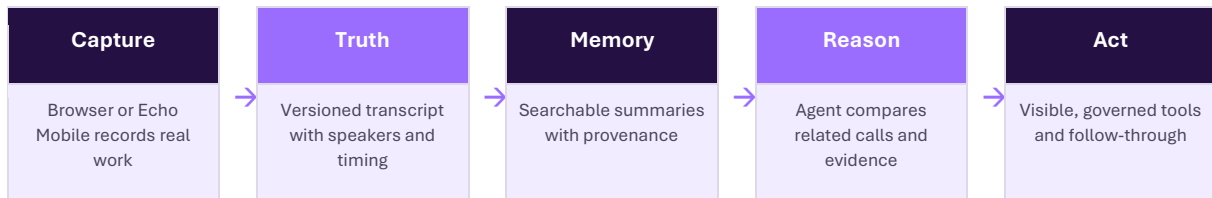
Old software	AI-native NeurAI Platform
The person navigates to a feature.	The assistant is present in the working surface.
The person retypes context into a prompt.	Context is attached from governed organizational memory.
Automation is a hidden workflow.	Actions are visible tool calls with consent and audit.
Permission is checked at the screen.	Authority is enforced at API, database role, row policy, grant, and tool policy.
Knowledge fragments across apps.	Echo creates the first high-fidelity memory substrate for future apps.

This is pioneering not because it adds a chatbot to a dashboard, but because the platform treats memory, authority, and action as one product system.

02 · WHY ECHO FIRST

The wedge is a conversation; the asset is organizational memory

Calls are frequent, information-dense, expensive to forget, and naturally connected to future action.



- Immediate value: searchable transcripts, summaries, corrections, speaker identity, and record management solve a clear daily problem.
- Compounding value: every processed call makes the assistant more context-rich for the next call, decision, and workflow.
- Trust laboratory: recording consent, sensitive content, tenancy, deletion, provenance, and hallucination risk force the platform to solve governance early.
- Platform proof: Echo exercises authentication, BFF boundaries, queues, storage, ML, LLM tools, connectors, administration, and responsive UX in one coherent app.
- Expansion bridge: once conversations become durable memory, the same substrate can power project, knowledge, workflow, and service apps without rebuilding identity or trust.

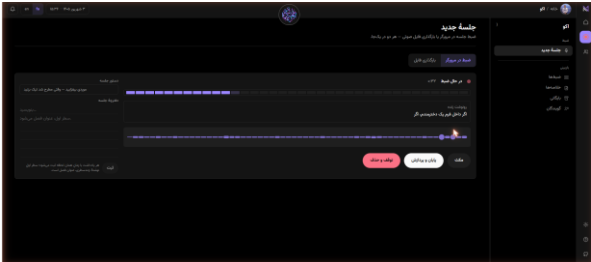
Strategic choice A horizontal assistant without a truth source is generic. A point meeting-notes tool without a platform is bounded. Echo gives NeurAI Platform both a decisive starting product and the memory layer from which a broader AI-native workspace can grow.

One calm shell; many specialized capabilities

The user experiences a single assistant platform, while Echo provides the first deep domain surface.



Persian-first platform hub: prompt, sources, modes, and the Echo app entry.



Echo capture: waveform, live transcript, agenda, notes, and visible recording controls.

Surface	What it does	Why it matters
Assistant hub	Ask across permitted conversations, attach sources, select agents/models, use web search when enabled.	Makes organizational memory conversational without separating it from governance.
Echo capture	Create a meeting, choose language/template, record, pause/resume, show live captions, buffer parts.	Reduces loss at the moment truth is created.
Records	Filter, bulk act, archive, restore, scope, inspect processing state.	Turns recordings into a manageable organizational corpus.
Record page	Audio, transcript, speakers, notes, versioned summaries, translation/edit controls, assistant pane.	Keeps source truth and derived insight visibly connected.
Speakers directory	Enroll voice samples, merge people, group/read speaker history.	Makes recurring participants recognizable without pretending uncertain labels are facts.
Skills / agents / workflows	Configure instructions, models, tools, reusable agent cards and workflows.	Moves from one generic assistant to governed, repeatable expertise.
Connectors / gateway	API keys, webhooks, connector accounts and secret storage.	Lets memory safely enter the rest of the organization's toolchain.
Management / platform console	Members, roles, models, server health, roots, org bootstrap and audits.	Makes a multi-tenant system operable instead of merely demoable.

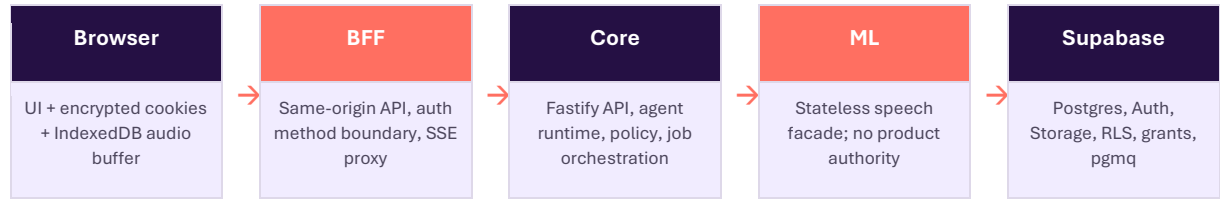
04 · FOUR PARTS, THREE PLANES

A small number of explicit boundaries

The platform is deliberately divided where trust, scale, and failure modes differ.

Experience plane	Next.js web · Persian/English shell · responsive recorder · assistant surfaces
Control plane	Fastify API · BFF boundary · agent runtime · workers · policy veto · audit
Data plane	Supabase Postgres/Auth/Storage · RLS · grants · pgmq · provenance

The four deployable parts are web/, core/, ml/, and Supabase. They form three planes: experience, control, and data. This is not microservice theatre: the ML boundary exists because speech dependencies and stateless media processing are operationally different; the BFF exists because browser credentials and internal topology should not leak; the worker exists because durable jobs have different lifecycle semantics from synchronous API requests.

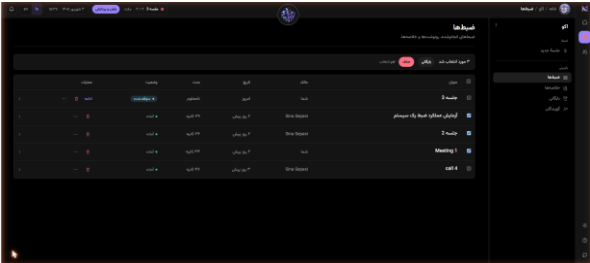


Why this shape Each boundary names a kind of authority. The browser holds a session, not provider secrets. ML receives a short-lived signed media URL, not product credentials. Workers re-resolve the record owner. Agent tools use the agent database role. The database remains the final wall.

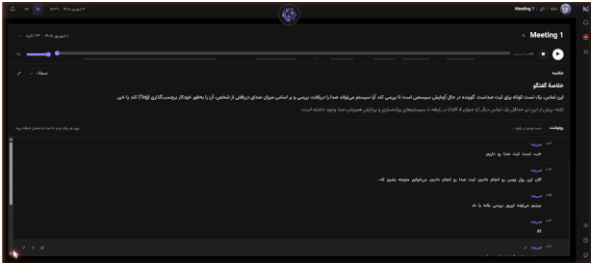
The experience plane is also a security boundary

Next.js renders the Persian-first shell while its BFF keeps tokens and internal services off the public edge.

Component	Responsibility	Why chosen / trade-off
Next.js App Router	Localized pages, server/client composition, route-level loading and error boundaries.	A cohesive React full-stack shell with strong deployment ergonomics; requires discipline around server/client boundaries.
BFF (Backend for Frontend)	Same-origin /api routes validate browser-facing input, manage auth cookies, proxy Core and SSE.	Reduces exposed topology and prevents direct browser possession of internal credentials. It adds one hop, paid back in control.
next-intl	Persian and English message catalogs, localized routing.	Makes locale a structural dimension rather than scattered string conditionals.
Tailwind + shared scaffold	Consistent tokens, menus, panels, RTL-aware spacing and responsive shell.	Fast iteration with one source of layout truth; rendered checks remain essential because CSS intent can differ from computed output.
IndexedDB capture buffer	Keeps audio parts resilient before upload/finish.	Browser crashes and network loss should not erase the meeting; durable local buffering is worth the extra state machine.
SSE	Streams assistant text/tool lifecycle and live captions downstream.	Works through the BFF and Vercel constraints; simpler than a browser-to-Core WebSocket for this topology.



Records management: filters, selection, batch actions, and status visibility.



Record detail: transcript source, audio, summary, notes, and contextual assistant.

One control plane, two execution tempos

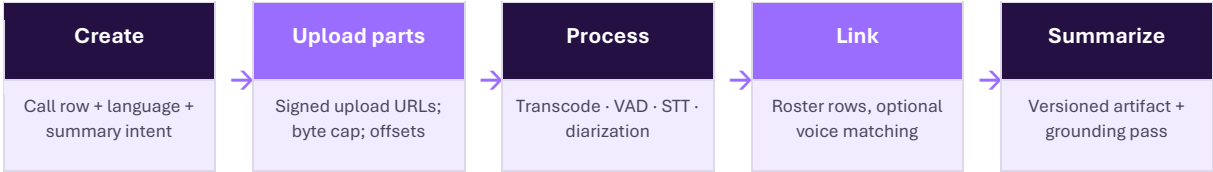
Fast synchronous request handling and durable asynchronous processing share identity rules but not failure semantics.

Unit	What it owns	Importance
Fastify API	Calls, parts, auth, assistant sessions, search, skills, agents, workflows, connectors, member/admin/platform routes.	A narrow, typed control surface that makes authorization and errors explicit.
Identity resolver	JWT sub → active app user → org, role, status; worker payload → call owner.	There is no useful database handle until a person is known.
Lifecycle repositories	Call/part state ladders, retry, missing-part handling, duration recompute, version pointers.	Makes progress and recovery durable, visible, and idempotent.
pgmq worker	Consumes process, speaker-link, summary, webhook, and signal jobs with retry/dead-letter behavior.	Meeting processing must survive request lifetimes and machine restarts.
Agent runtime	One Pi-based runtime for assistant and summarizer, with different tools and prompts.	Avoids two drifting intelligence stacks and preserves run-level audit/replay.
Gateway	Scoped API keys, signed webhooks, replay tolerance, redirect/SSRF controls.	Connects the platform outward without weakening the inbound trust model.

The worker runs as the owner A background job does not become a superuser because no browser is present. The payload carries the owner identity; Core re-resolves it and performs product work through the same RLS-bound door.

The transcript is truth; everything else can be rebuilt

The pipeline preserves recoverability, provenance, and honest degradation at every stage.

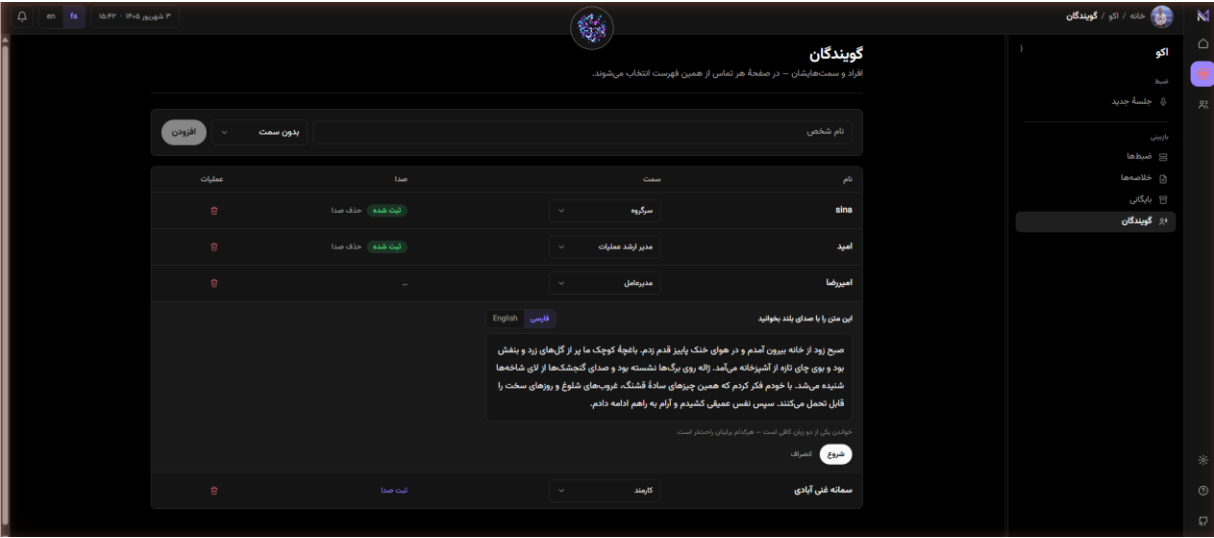


Principle	Implementation consequence
Transcript is source of truth	Transcript segments and audio provenance outlive summary versions; a summary is derived and rebuildable.
Idempotency keys the artifact	The worker checks for stored transcript rows rather than trusting a possibly stale status flag.
Missing is a state	A failed/missing part can be marked explicitly, allowing the rest of a call to settle without inventing continuity.
Derived artifacts are versioned	Human summary edits create new versions; the current pointer advances without erasing history.
Delete is role-aware and recoverable	Soft deletion hides immediately and allows a 30-day restore; hard purge is not an agent tool.
Provenance travels with output	Model, lane, timing granularity, diarization source, warnings, editor, and run records explain what produced the artifact.

Persian accuracy is a pipeline, not a checkbox

Echo separates capture, speech detection, transcription, speaker structure, and voice identity so each can degrade honestly.

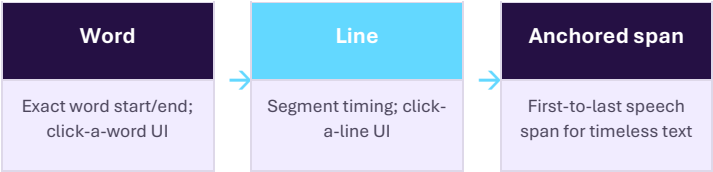
Stage	Current design	Why
Transcode / probe	ffmpeg normalizes media; duration/channel facts are measured.	Provider inputs become predictable and oversize media fails before spend.
VAD	Local speech detection removes meaningful silence; no-speech is logged and the original still reaches STT.	Lower spend and latency without silently trusting an all-negative detector.
STT lanes	Soniox primary, OpenRouter fallback; Persian/English hints and org glossary bias.	Cloud accuracy with explicit fallback and user language steering.
Channel logic	True two-channel audio maps speakers by microphone; dual-mono is detected and downmixed.	Prevents duplicated words, invented speakers, and double billing.
Diarization	Local sherpa-onnx CPU path for mono when needed; provider or channel labels otherwise.	Speech structure stays productless and optional while avoiding a mandatory GPU service.
Voice identity	Enrollment embeddings and conservative matching link call speakers to directory people.	Recurring participants become useful context without claiming uncertain matches as fact.
Live captions	Core holds the provider WebSocket; browser uploads audio and receives caption events via an ephemeral owned/ticketed session.	Provider keys never enter the browser; captions remain content and never logs.



Voice enrollment and speaker directory: identity is cultivated as a governed asset, not guessed from a filename.

When fidelity falls, honesty remains

A transcript must stay seekable even if a provider cannot return word-level timestamps.



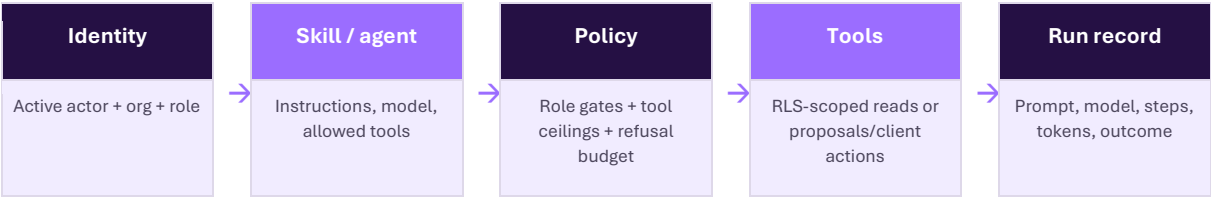
ML returns timings on the processed file’s zero-based timeline. Core anchors them to each call part’s offset. If a provider returns prose without time, Echo synthesizes one honest speech-span segment rather than zeros or an unseekable void. Non-empty segments with $end \leq start$ are refused.

Why it matters Silently losing time makes a correct transcript feel broken and destroys citation. The timing ladder turns provider variation into an explicit product mode instead of a hidden defect.

Granularity	What the user can do	What provenance says
Word	Seek and highlight at the word level.	timestamps: word; has_word_timestamps: true
Segment	Seek to a sentence/line span.	timestamps: segment; feature demotes honestly
None → anchored speech span	Seek to the coarse speech region and read text.	timestamps: none; degraded: true; real non-zero span

Assistant and summarizer share the same governed intelligence

Different jobs receive different prompts and tools, but every run is identity-bound, recorded, budgeted, and replayable.



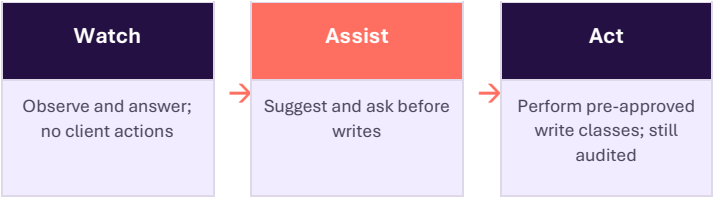
Agent capability	Mechanism	Guardrail
Read organizational memory	search_transcripts, read_window, get_call, list_related_calls	Identity wrapper + RLS; cited call/time evidence.
Propose content changes	correct_transcript, edit_speaker_roster, replace_summary	Proposal first; explicit human confirmation applies through the app role.
Operate the visible UI	navigation, capture controls, record/member/directory actions	Only tools advertised by the surface; consent flags follow the autonomy setting.
Search the web	OpenRouter online variant when the user enables it	Base model passes catalogue rules before transport suffixing.
Summarize a call	Same runtime with summarizer skill, related-call tools, templates, and grounding verifier	Runs as the call owner; an empty result or missing system floor fails loudly.

The profound choice The agent does not have its own omniscient identity. It borrows the caller’s authority and never more. That makes AI a participant inside the security model—not an exception around it.

From chat feature to operating surface

The platform’s AI-native layers answer four questions: where is the assistant, what can it touch, what does it notice, and when may it act?

Layer	Meaning	Current architectural expression
Presence	Assistant available in the hub and contextual panels.	Shared session system, call context chips, visible tool lifecycle, persistent conversation history.
Hands	Tools that read memory or operate the product surface.	Server domain tools, propose-only write tools, and client-executed UI/write tools.
Signals	Events that can create a brief, digest, or rule-triggered follow-up.	call.processed, weekly digest and agent-rule queue patterns; event identity preserved.
Autonomy	Watch → Assist → Act controls consequence, not intelligence.	Watch offers no client tools; Assist requests consent for writes; Act is reserved for approved write classes.



Autonomy is a policy dimension, not a personality setting. The same model can be helpful at every rung; what changes is which effects are exposed and when a person must consent.

Prompts guide; database walls enforce

Sensitive conversation data requires security properties that remain true even when application code or a model is wrong.

Layer	Enforcement	Failure it prevents
JWT verification	Signed access token verified against trusted JWKS; sub becomes identity candidate.	Forged or expired browser identity.
Active user resolution	App user, org, role, status checked before work.	Pending, disabled, suspended, or unknown identities becoming product actors.
Identity-bound DB factory	SET LOCAL ROLE + set_config(actor_id) inside the same transaction.	Pooled connections leaking authority or code querying without a caller.
RLS	Every protected row is filtered by actor/org/ownership/share policies.	Cross-organization and cross-owner access.
Role grants	echo_app and narrower echo_agent grants; agent has no DELETE.	A model or tool gaining authority through a generous query path.
Central tool veto	Declared-tool filter, admin gates, total/blocked attempt budgets.	Tool discovery drift, permission grinding, and unbounded loops.
Proposal + consent	Consequential content changes are proposed before app-role application.	A plausible but wrong model edit becoming truth.
Audit/provenance	Agent runs, steps, versions, editors, webhook delivery records.	Invisible AI behavior and irreproducible derived artifacts.

Non-negotiable invariant No database access without user identity. The agent borrows the caller’s authority. RLS plus grants are the wall. The agent role has no DELETE. Content never enters logs.

A record of truth, derivatives, actors, and decisions

The schema is handwritten SQL because the wall must be reviewable at the level where it is enforced.

Domain cluster	Key entities	Why they exist
Tenancy & identity	org, app_user, invitation, status history, platform_operator	One-org v1 membership, explicit roles/status, bootstrap and root audit.
Conversation truth	call, call_part, transcript_segment, call_speaker, person	Durable audio/transcript timeline, speaker roster, directory identity.
Derived intelligence	summary, skill, agent_run	Versioned summaries, reusable instructions, replayable model/tool execution.
Assistant	agent_session, agent_message, feedback, share	Persistent conversations with deletion semantics that preserve audit.
Automation	assistant_agent, workflow_template, agent_card, agent_rule	Named expertise, reusable process, proactive signals and autonomy.
Integration	api_key, webhook, delivery, connector_connection, connector_secret	External entry/exit with scoped authority and secret isolation.
Governance	admin_action, platform_audit, deletion_record, proposal_decision	Human and system accountability across sensitive actions.

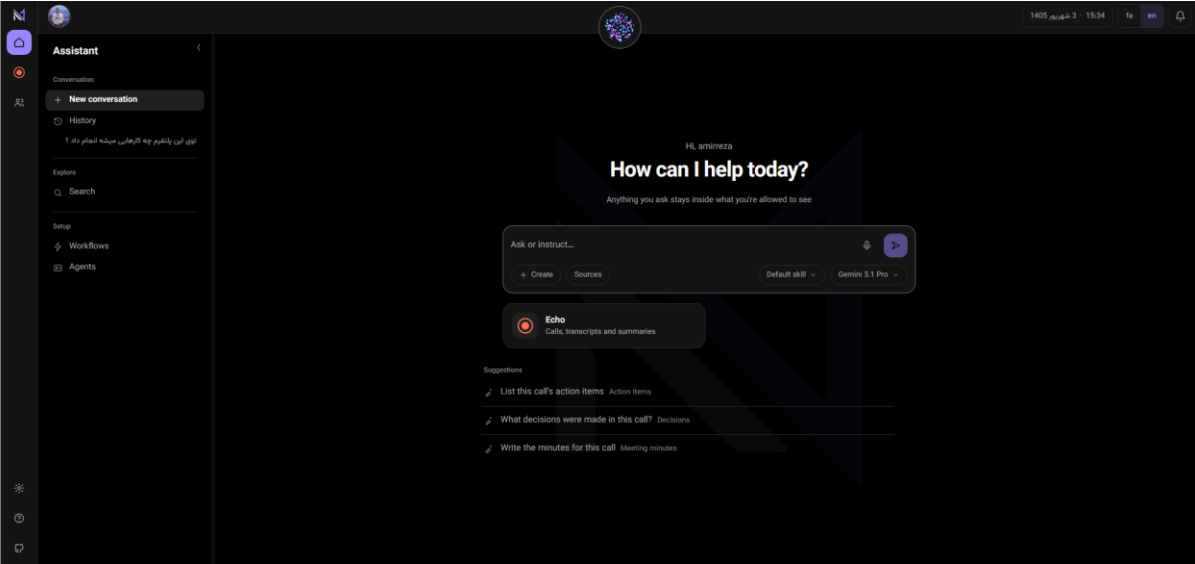
Drizzle is used for queries only where appropriate; schema and security are handwritten numbered migrations. RLS and grant changes ship with SQL tests because the critical contract lives in Postgres, not in TypeScript types.

RTL is only the visible layer

Language, search, speech, dates, digits, names, and layouts all need first-class Persian behavior.

Dimension	Design requirement	Why it matters
Direction	RTL layout, start/end spacing, mirrored navigation, scoped LTR for ids/timestamps.	A translated LTR shell still feels foreign and breaks mixed-script details.
Normalization	Normalize Persian/Arabic character variants at ingest and query.	Search should not miss the same word because of ی/ی or ک/ک variants.
Digits & dates	Persian digits and Jalali-capable presentation with user preference.	Dates and numbers are product semantics, not decoration.
Speech	fa/en hints, mixed-language handling, glossary bias for names and org vocabulary.	Proper names and code-switching are where generic transcription fails visibly.
Typography	Persian-capable fonts and line-height tuned for Arabic script.	Readable text density and diacritics require different visual rhythm.
Testing	Sweep the structurally less-viewed locale and verify rendered state.	A complete fa.json can hide a broken en.json—and the inverse.

«نمونه تجربه»: خلاصه جلسه را با منبع و زمان نشان بده



The same platform shell in English: locale is a first-class route and message-catalog dimension, not an afterthought.

Memory becomes valuable when it can travel safely

The gateway is the platform’s contract with external systems; connectors are governed accounts, not copied secrets.

Mechanism	How it works	Trust property
Scoped API keys	Admin-created credentials with scopes, prefix/hash storage, last-used and revoke lifecycle.	The raw key is shown once and cannot be recovered from the database.
Webhooks	Subscribed events, signed deliveries, timestamped HMAC, replay tolerance, retries and delivery history.	Consumers can verify origin; redirects and targets are re-checked against SSRF rules.
Connector accounts	Org-scoped connection metadata; secret material stored through a dedicated secret backend.	Product rows can describe a connection without exposing provider tokens.
OAuth / PKCE	Provider authorization can bind a person or org without sharing passwords.	Authorization code interception and browser secret exposure are reduced.
Workflow tools	Agents/workflows select from approved tools and connector capabilities.	An integration broadens utility without broadening the caller’s authority.

Adapter rule Providers spell absence, errors, and retries differently. The adapter decides what 'already absent' means and returns a typed state; callers must not infer provider semantics from a raw status code.

16 · DEPLOYMENT & OPERATIONS

Managed edges, self-controlled compute, one observable system

The deployment follows the risk boundary: public UI at the edge, controlled Core/ML compute, managed durable data.

Layer	Placement	Reason
Web / BFF	Vercel	Global delivery, same-origin browser boundary, managed build/deploy path.
Core API + worker	Hetzner	Long-lived processes, queue consumers, provider relays, explicit service supervision.
ML	Hetzner CPU service	Local ONNX/VAD/ffmpeg dependencies near Core without placing product state in ML.
Private ingress	Cloudflare Tunnel	No public origin exposure for internal services; managed edge routing.
Data/Auth/Storage	Managed Supabase	Postgres/RLS, authentication, object storage, and operational durability in one data plane.
Models	Soniox + OpenRouter	Specialized speech and user-selectable LLM lanes behind server-held keys.

- Health checks resolve the specific callable they guard; module presence is not health.
- Production-runtime boot tests start each process and make it answer one request.
- Logs carry codes, identifiers, counts, and structured database fields—never transcript content or signed URLs.
- Live network/spend tests are opt-in, run at package acceptance, and repeated at release gates.
- Runbooks separate deployed health from local code presence; a repository route is not proof that the external environment is healthy.

A glossary for the whole platform

The platform is sophisticated; its explanation should not require a security or ML background.

Term	Plain-language explanation
BFF	Backend for Frontend. A server layer built specifically for the web UI. It holds secure cookies, reshapes requests, and proxies Core so the browser does not talk directly to internal services.
JWT	JSON Web Token. A signed, time-limited identity claim. Echo verifies the signature and reads the subject (sub); it does not trust a user id supplied in a request body.
JWKS	JSON Web Key Set. The trusted public keys used to verify JWT signatures and key rotation.
RLS	Row-Level Security. PostgreSQL policies that decide which rows the current actor may see or change—even if an application query forgets a filter.
RBAC	Role-Based Access Control. Owner, admin, and member define broad authority; RLS and ownership add record-level precision.
Database grants	Permissions attached to database roles. They are the hard ceiling: echo_agent has no DELETE even if a prompt or tool tries.
SSE	Server-Sent Events. A one-way streaming HTTP connection used for assistant text, tool lifecycle, and caption events.
WebSocket	A two-way long-lived connection. Core uses one to the live STT provider; the browser does not receive the provider key.
PKCE	Proof Key for Code Exchange. An OAuth protection that ties the callback code to the browser session that started authorization.
HMAC	A keyed signature over webhook content and time. Receivers can verify a delivery came from Echo and was not replayed outside the tolerance window.
SSRF	Server-Side Request Forgery. An attack that tricks a server into calling internal/private addresses; webhook targets and redirects are checked to prevent it.
STT	Speech-to-Text. The model/provider lane that turns audio into words, timing, language and sometimes speaker labels.
VAD	Voice Activity Detection. A local model that identifies speech regions so long silence can be removed before paid transcription.
Diarization	Answering ‘who spoke when?’ by clustering one audio stream into speaker labels such as S1 and S2. It is not the same as knowing the person’s name.
Voiceprint	An embedding derived from voice samples and compared conservatively to recurring speakers. It supports a suggested match, not infallible identity.
Embedding	A numeric vector representing features of speech or text. Similar vectors can indicate related voices or meaning.
LLM	Large Language Model. Generates or interprets language; in NeurAI Platform it is surrounded by tools, identity, policy, provenance, and failure checks.
Tool calling	The model asks software to perform a named operation with structured parameters. The platform still decides whether that tool is allowed and how it is executed.
Prompt injection	Instructions hidden in retrieved content that try to redirect an agent. Echo quotes transcript content as data and enforces permission outside the prompt.

Term	Plain-language explanation
Provenance	The record of what produced an artifact: provider/model, lane, timing mode, run, editor, warnings, and source.
Idempotency	A retry produces one durable outcome, not duplicates. The worker checks the artifact that would prove prior success.
pgmq	A durable Postgres-backed message queue used for background processing and retries.
DLQ	Dead-letter queue. Jobs that exhausted safe retries land here for diagnosis instead of disappearing or looping forever.
Signed URL	A short-lived capability to upload or download one object. ML can read the media it needs without holding platform storage credentials.
Soft delete	Hide a record immediately but retain it for a recovery window; Echo uses a 30-day restore period before purge.
Tombstone	A minimal durable record that an object was deleted, useful for audit and synchronization without retaining the original content.
Bidi / RTL	Bidirectional and right-to-left layout behavior. Persian UI must place content and controls by start/end semantics while keeping ids and timestamps readable.
Jalali calendar	The Persian calendar system supported as a display preference; storage remains unambiguous machine time.

The category is proven; the opening is still distinct

Gong, Otter, and Fireflies validate demand for conversation intelligence—NeurAI Platform competes on Persian-first depth, authority-aware action, and a broader platform thesis.

Platform	Best-known strength	Closest overlap	NeurAI Platform’s strategic contrast
Gong	Revenue-focused conversation intelligence, deal signals, coaching, forecasting, embedded revenue agents.	Capture, transcription, cross-conversation insight, action from interaction data.	NeurAI is not limited to the revenue operating model: Echo is the first memory app inside a general AI-native platform, with caller-bound authority as a core primitive.
Otter.ai	General meeting capture, live transcription, summaries, AI chat, enterprise workspace administration.	Meeting knowledge, searchable transcripts, assistant Q&A, multi-device capture.	Otter’s official May 2026 language list does not include Persian; NeurAI is Persian-first across UI, search, speech hints, dates, and organizational vocabulary.
Fireflies.ai	Broad meeting automation, AskFred Q&A, AI skills, conversation intelligence, extensive integrations.	Custom summaries/skills, cross-meeting questions, workflow/integration ecosystem.	NeurAI makes product actions and data access flow through the same identity/RLS/grant wall, and treats autonomy as a visible Watch→Assist→Act policy.

How to compete Do not try to out-Gong Gong on revenue dashboards or out-integrate Fireflies on day one. Win the underserved Persian-first trust surface, make the agent visibly useful inside the product, and let Echo’s governed memory become the substrate for multiple apps.

This comparison reflects public positioning as of 26 August 2026. It is a strategic comparison, not an independent benchmark of accuracy, security certifications, or total feature coverage.

Official competitor sources: [Gong Conversation Intelligence](#) · [Otter home / Conversational Knowledge Engine](#) · [Otter supported transcription languages](#) · [Fireflies platform](#) · [Fireflies AskFred API](#)

19 · CHOICE RECORD

Why these technologies—and what was consciously rejected

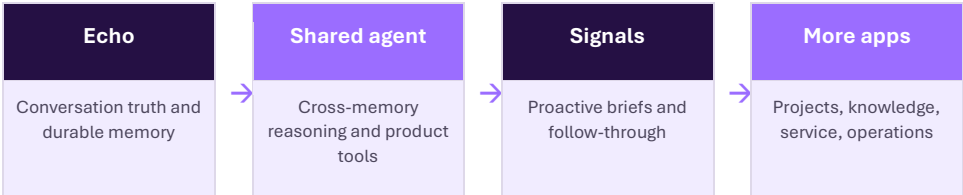
The architecture is strongest where it records the trade-off instead of hiding it.

Choice	Selected	Why	Rejected / deferred
Web architecture	Next.js + BFF	One localized shell and secure same-origin boundary.	Browser → Core direct calls; exposes internal topology and complicates auth.
API	Fastify + Zod	Typed, small, fast control surface with explicit validation.	Framework-heavy service mesh before the domain needs it.
Database	Supabase Postgres + handwritten SQL	RLS/grants/schema visible and testable where enforced.	ORM-owned schema that obscures the security wall.
Jobs	pqmq	Durable queue near transactional data with operational simplicity.	In-memory jobs; they disappear at restart. Separate broker can come when scale proves it.
Agent runtime	Pi + central policy + typed tools	One runtime for assistant/summarizer with tool-call interception and replay.	Prompt-only permission; two independent agent stacks.
Speech	Soniox + OpenRouter fallback + local ONNX	Persian-capable cloud STT with stateless local speech structure.	Browser provider keys; a stateful product-aware ML service.
Live transport	Chunk upload + SSE / ephemeral ticket	Works with BFF/Vercel topology while Core owns provider socket.	Browser-to-provider WebSocket containing provider credential.
Delete	Soft delete + named purge doors	Recoverability, role hierarchy, audit, and no agent DELETE.	Generic DELETE permission or hidden service-account bypass.

Build the substrate once; compound it through apps

Echo proves the hard platform capabilities. Future apps should reuse identity, memory, tools, connectors, and governance—not fork them.

Now: Echo + shared platform	Next: deepen	Then: expand
Capture, transcript, speaker structure, versioned summaries, search, assistant sessions, skills/agents/workflows, connectors/gateway, management and platform-root surfaces.	Acceptance-grade Persian speech quality, stronger templates/grounding, operational SLOs, richer signals, approved Act classes, production connector catalogue.	Project memory, knowledge/wiki, service workflows, org briefs, cross-app actions—all on the same identity and memory substrate.



Expansion rule A new app is valuable when it contributes a new source of truth or a new governed action surface. It should not create a second identity system, a second memory store, or an unbounded agent.

Build the AI people can trust with the work that matters

NeurAI Platform starts with the most human data in an organization—conversation—and turns it into memory without giving up authority, provenance, or language.

Echo is the first app because it transforms an urgent loss—forgotten conversations—into a compounding asset. NeurAI Platform is the larger bet: when every memory is attributable, every action is bounded, and every interface is built for the person's language, AI stops being a side chat and becomes a trustworthy operating surface.

The promise Capture what happened. Explain what it means. Act only with authority. Learn across time. Do it naturally in Persian. Then let every future app inherit that trust.

That is why the platform begins with Echo—and why Echo is much more than meeting notes.

APPENDIX · SOURCE REGISTER

What this edition is grounded in

Primary project sources first; official competitor sources only for market positioning.

Source	Role
docs/SPEC.md	Product behavior source of truth.
ARCHITECTURE.md	Locked implementation decisions M1–M40 and amendments.
docs/PLATFORM-BRIEF.md	Platform narrative, app model, and strategic framing.
docs/AI-NATIVE-PLAN.md	Presence, hands, signals, autonomy and sequence.
docs/CONNECTORS.md	Connector boundary and product vocabulary.
docs/PLATFORM-ROOT.md	Bootstrap/root administration behavior.
docs/PLATFORM-OPERATIONS-RUNBOOK.md	Deployment and operational practices.
docs/CLOSE-m4-frontend.md	Frontend closure evidence and UI implementation notes.
docs/screenshots/*	Current visual evidence of the web product surfaces.
web/, core/, ml/, db/	Route, dependency, runtime, schema, security and current capability evidence.
Gong / Otter / Fireflies official product and help pages	Competitive public positioning as of 26 August 2026.

Important: repository evidence is not a live production probe. Where the code contains a capability but external provider credentials, migrations, models, or deployed versions determine availability, this document describes the architecture/current implementation rather than claiming a verified customer-facing service level.